CHAPTER IV
Functionalism

It was in the middle of the twentieth century that the concept of functionalism evolved as an alternative to the mind–brain identity theory and behaviourism. Today, functionalism is widely regarded as one of the most influential views in the entire field of philosophy of mind. As opposed to characterising mental states based only on their physical makeup or on the actions of the observer alone, functionalists define mental states in terms of the roles that they play within a system. This is in contrast to the traditional practice of describing mental states. It is possible to get an understanding of a mental state by witnessing its behaviours, which includes how the condition is brought about by inputs, how it interacts with other mental states, and how it results in behavioural outputs. A great number of philosophers played a significant role in the development and refining of functionalism, which in turn affected contemporary debates in the domains of philosophy, cognitive science, and artificial intelligence as a result of their contributions.

Hilary Putnam who is considered to be one of the founding figures of functionalism. Putnam introduced the idea of machine-state functionalism in the 1960s. This proposed that the human mind is analogous to a computer program, and that the brain is analogous to hardware. Putnam maintains that mental states are comparable to computational states in the sense that, rather of being defined by their physical material, they are dependent on the functional interactions that exist between them. The theory of multiple realisability, which asserts that the same mental state can be realised in other physical systems, such as human brains, animal nervous systems, or even artificial devices, is the most influential contribution that he has made. This idea was developed by him. The identity hypothesis, which argued that certain mental states are similar to specific brain states, was open to a direct challenge as a result of this idea. As part of the process of establishing functionalism as a powerful and adaptable account of the mind, Putnam's work played a significant role.

Jerry A. Fodor is another renowned functionalist philosopher who is credited with bringing together the functionalist philosophy and a strong commitment to mental representation into a coherent philosophical framework. Fodor was the one who came up with the Language of Thought (LOT) theoretical framework. An internal symbolic language that is directed by syntactic rules is said to be the medium through which thinking takes place, according to this

hypothesis. According to Fodor, mental states are functional states that involve representations that hold semantic information. These representations carry out mental processes. His interpretation of functionalism, which placed a focus on the computational side of cognition, had a tremendous impact on the field of cognitive research. His interpretation of functionalism was particularly influential. In order to better understand the manner in which mental states are accountable for behaviour, Fodor argued that functional explanations are necessary. This, in turn, strengthened the capacity of functionalism to provide an explanation for behaviour.

It is generally agreed that David Lewis was the one who first developed the method of psychology that is today known as analytic functionalism. Regular psychology describes mental states in terms of the causal functions that they play, and this method describes mental states in terms of those functions. The position of ideas such as "belief," "desire," and "pain" within a network of relations that connect perception, action, and a variety of other mental processes is what Lewis considers to be the defining of these notions. Following the description of these responsibilities, it is feasible to determine the states that occupy these roles by comparing them to their appropriate physical states. By proving that mental notions can be expressed functionally without rejecting their physical manifestation, Lewis's approach was an attempt to bring functionalism and physicalism into harmony with one another. This was by demonstrating that the two schools of thought are compatible with one another.

Daniel Dennett offered a pragmatic and convincing defence of the functionalism philosophical school of thinking by contributing his concept of the intentional posture to the discussion. Dennett asserted that a suitable way for predicting the behaviour of a system is to assign the system's beliefs, goals, and intentions to the system itself. This holds true regardless of whether or not these states exist as private entities that are located within themselves. In accordance with this point of view, mental states are patterns that are able to be defined at the level of functionality. Because Dennett's work extends the framework of functionalism beyond the sphere of humans and into the realms of animals, machines, and artificial intelligence, it poses a challenge to the conventional concepts of consciousness and mentality that have been prevalent in the past.

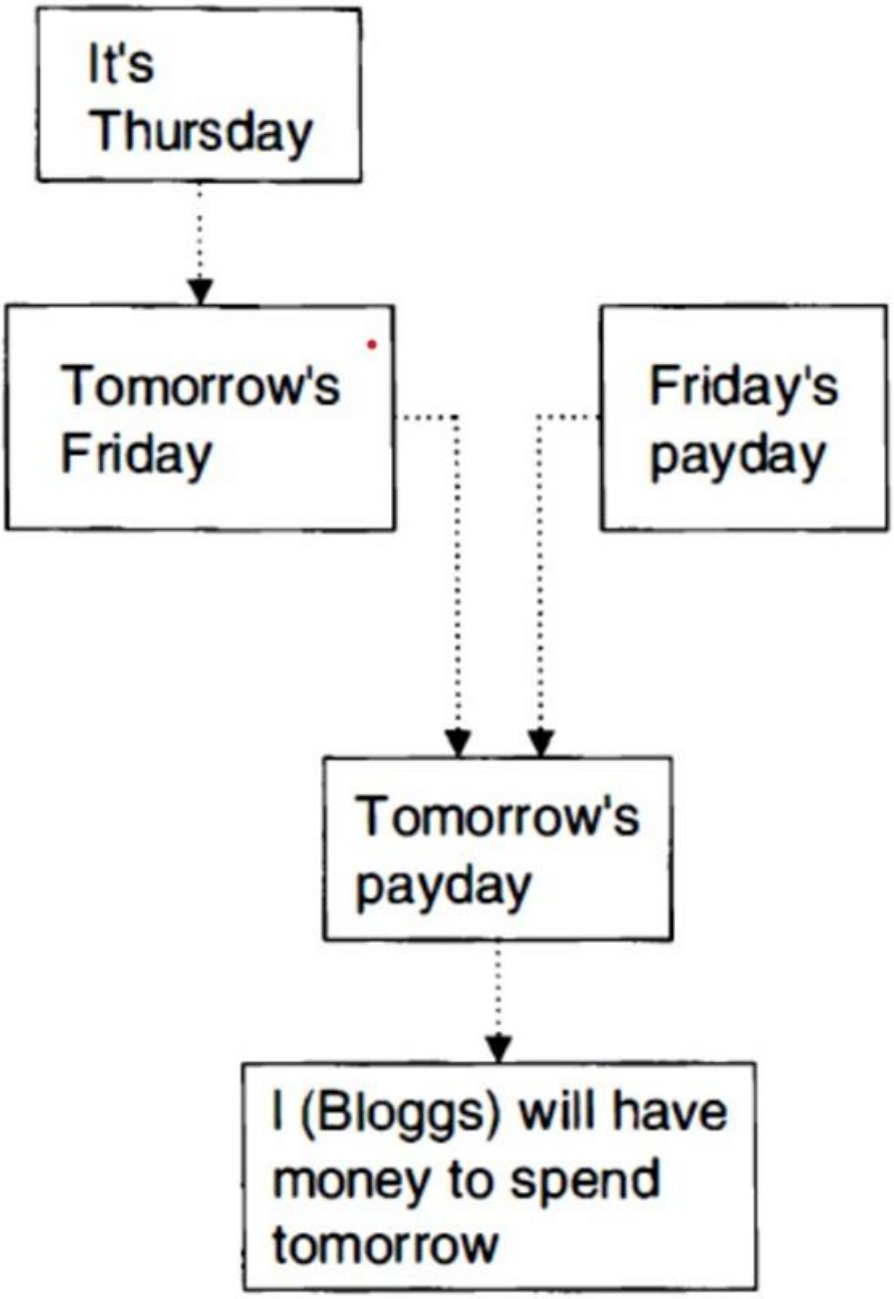
An additional notable contributor to this topic is Sydney Shoemaker, who, within the context of a functionalist framework, focused on mental causation and self-knowledge. On the other hand, Shoemaker contended that mental states must have actual causal capacities, and that functionalism may be able to account for this without reducing mental qualities to basic behavioural dispositions. In addition to this, he claimed that this is entirely consistent with the objective of functionalism. He was able to address criticisms that functionalism ignores

subjective experience as a result of his work, which contributed to the improvement of the logical foundations of the concept of functionalism. The functionalist philosophical perspective asserts that mental states are determined by the causal interactions that they have with one another, as well as by the sensory inputs and behavioural outputs that they experience. This is in addition to the fact that mental states are defined by individual experiences.

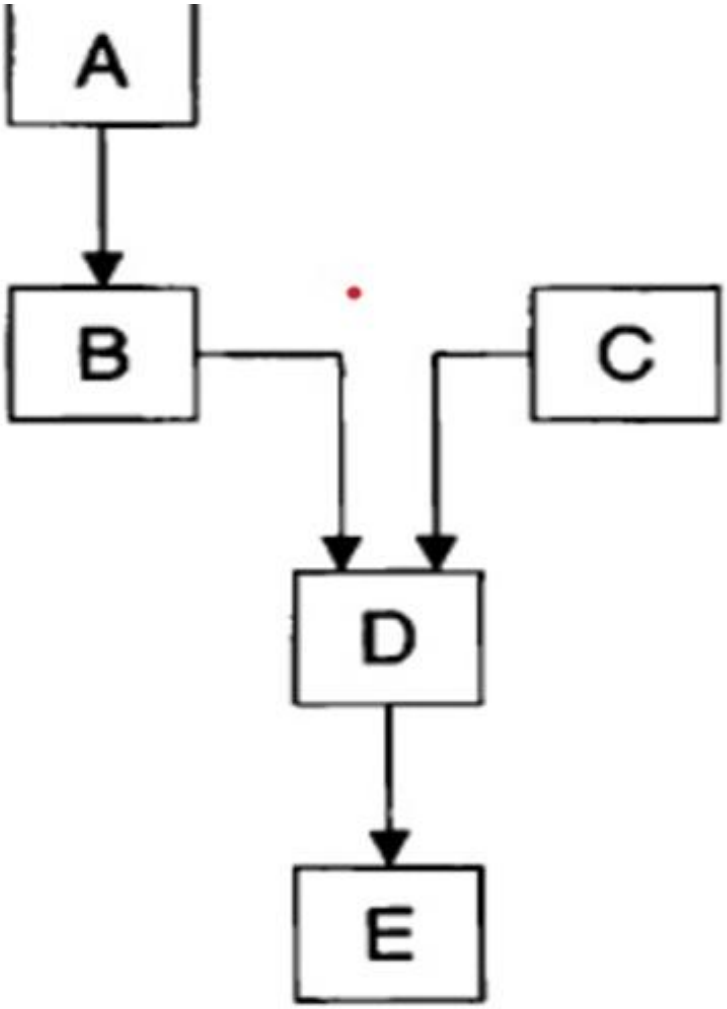
A significant theoretical accomplishment in the field of analytic philosophy during the entirety of the twentieth century is the functionalist school of thought. In addition to this, it is the conceptual foundation upon which a significant amount of research in the field of cognitive science is built. A functionalist approach to philosophy is based on the concept that many entities can be adequately defined not by their composition but by the functions that they perform. This is the theoretical foundation of the functionalist method. The capacity of the knife to cut is what differentiates it from other types of knives; the difference between steel and plastic is immaterial to the classification of the knife as a knife. This is true regardless of the material composition of the knife. A similar assumption is relevant to mental states and processes, according to the functionalist, who thinks that this is the case. It is not the material components of a given mental state, such as brain activity, that constitute its essence; rather, it is the functions that it involves and the kinds of causes and effects that it generates that establish its essence. In the theory of functionalism, it is proposed that mental states are distinguished by the functions that they normally perform, which are sometimes referred to as their "functional" roles. The viewpoint of a comprehensive functionalist is to assert that the causal function of a mental state (M) is the factor that determines all of the qualities of M. This is the position that all comprehensive functionalists support. In light of this, the causal function that M possesses is what determines the content of M. Using a functional role theory of content, one can gain an understanding of the process that is responsible for the establishment of causal roles. It is common practice to carry out the process of determining functionalist role theories of content in this manner. As a general rule, move forward. One have to take into consideration the fact that the mental states of an individual normally produce a network of causal relationships or connections. Take for instance the standard approach to the situation. For instance, Bloggs's conviction that it is Thursday led him to the conclusion that tomorrow is Friday; his belief that tomorrow is Friday, in conjunction with his belief that Friday is payday, resulted in his assumption that tomorrow is payday; and his belief that tomorrow is payday caused him to believe that he will have money to spend over the course of the following day. There is, in general, a causal network that contains mental states, and there is also an inferential

network that encompasses the substance of those beliefs. Both of these networks are examples of networks. At this point, we have merely acknowledged the existence of two networks: a causal network that connects mental states and an inferential network that associates the contents of mental states. Both of these networks are inferential networks. The substance of Bloggs' ideas in the example is presented as a cohesive argument, proceeding from a sequence of premises to the conclusion, which is that "I (Bloggs) will only need to spend tomorrow." It is crucial to observe that this line of reasoning is presented in a way that is consistent with the argument. It is the contention of those who subscribe to functional role theories of content that individuals attribute content to mental states by creating a correlation between the inferential network and the causal network.

An inferential network



A causal network involving five beliefs (A-E)



The scenario that is associated with the perception of pain is characterised by activation of the C-fiber in humans, whereas in squids, it is indicated by activation of the D-fiber. In humans, the C-fiber is activated when feeling pain. According to the functionalist hypothesis of the brain, it is believed that the activation of C-fibres in humans serves the same role as the activity of D-fibres in Flipper. According to this point of view, the presence of an internal state that

performs a specific purpose is what distinguishes the sensation of pain from other states of being. When an internal situation is the outcome of a bodily harm, it is said to be performing the "pain function." This function leads us to scream "ouch" and massage the area that is afflicted by the pain. It is the contention of the functionalist view that the experience of pain is a consequence of having an internal condition that is brought on by a physical harm. In response to this circumstance, we find ourselves uttering the phrase "ouch" and massaging the region that is being impacted. Putting this into a broader framework, the theory of functionalism posits that in order to have mental state M, one must also have an internal condition that satisfies the 'M-function.' This is a requirement for having mental state M.

Antibiotics are a class of chemical that eliminates bacteria that are harmful to the patient to the greatest extent feasible while causing the patient the least amount of harm possible. Penicillin is considered to be an antibiotic by medical specialists due to the fact that it removes hazardous microorganisms while inflicting the patient the least amount of harm possible. The antibiotic classification is given to erythromycin because of its capacity to eradicate pathogenic germs while inflicting the patient only a minimal amount of harm. These two antibiotics, penicillin and erythromycin, on the other hand, have chemical structures that are extremely unlike to one another..

When it comes to a number of significant aspects, the philosophical theory known as functionalism suggests that mental states are analogous to carburettors and antibiotics. The purpose of a carburettor is to carry out the "carburettor work," whereas the role of an antibiotic is to carry out the "antibiotic job." When compared to an antibiotic, a carburettor is defined by its function of carrying out. In a similar vein, the distinguishing characteristic of a certain mental state is the degree to which it is able to perform the functions that are associated with that state.

In the context of the concept of many realisations, it is an idea that suffering can present itself in a number of different ways. The following is a list of the numerous manifestations of pain, as viewed from the functionalist point of view. In the field of philosophy of mind, the term "functionalism" refers to a condition that takes on the role of suffering. In accordance with this idea, an organism is considered to be in a state of experiencing pain if it either possesses an internal state that is capable of performing the function of pain or if it is afflicted with a condition that takes on the function of pain.

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Inputs, outputs, and internal links are the distinctive qualities that are associated with the pain role. Instances such as walking on a tack, fracturing a leg, and boiling your hand are all instances of scenarios that have the potential to inflict significant pain. Inputs are the conditions that are responsible for causing discomfort. A person's outputs are the behaviours that are a direct result of experiencing pain. Some examples of outputs include, but are not limited to, shouting, rubbing the affected area, and vocalising "ouch." Pain and a range of mental states are represented by the internal connections, which are the causal interconnections between the two. The causal relationship that exists between pain and anxiety, for instance, states that acute pain often induces anxiety. This is one example of what they contain. Pain is one of the conditions that can be brought on by walking on a tack. Pain frequently brings about thoughts of dread and produces statements such as "ouch." These sensations can be brought on by walking on a tack.

Although functionalism accepts that it is impossible to directly correlate different varieties of mental states with different types of behavioural dispositions, it nonetheless strives to characterise mental states through behaviour, although in a roundabout fashion. According to the functionalist theory, the role that a mental state plays in the causal chain can be broken down into three distinct types of interactions that are deemed to be causal.

The way in which light that enters the eyes forms a visual impression is an example of how various characteristics of a subject's surroundings might first produce particular mental processes. This is because light is able to penetrate the eyes. Secondly, it is possible for certain mental states to have a causal interaction with other mental states that are present within the same individual. In order to illustrate this point, consider the fact that a feeling of discomfort in the leg may provide the impression that the limb in question has been harmed.

When it comes to the actions that an individual engages in with their body, certain mental states can have a causal influence on those activities. For instance, a person who feels that they have damaged their leg and subsequently has a great desire to alleviate the pain that they are experiencing is an example of this phenomenon. In the event that this occurs, the person may end up touching the injured area or removing their leg from a potentially dangerous situation.

According to functionalists, the categorisation of a mental state into a specific category is not fundamentally dependent on the mental state's underlying material makeup; rather, it is based on the mental state's functional role within the system to which it is attached. They believe that this is the most important factor in determining the classification of a mental state. The theory of functionalism is consistent with the idea that individuals and groups may exhibit sufficient

similarity in their functional organization to possess comparable mental states. This idea is consistent with the concept of functionalism. According to the functionalist theory, mental states are the bearers of a variety of causal functions. This is something that we have observed up to this point in our comprehension of the theory concerning mental states. Furthermore, we made the remark that, in principle, the roles that are inherent to various mental states can be completed by a wide variety of states. This was a significant finding for us. Because of this, functionalism is able to shed light on the infinite variety of mental states that are realisable. Now that we have arrived at this stage, we will study the link that exists between functionalism and the following theories: (i) type identity theory; (ii) restricted type identity theory; and (iii) token identity theory.

According to the functionalist hypothesis, the reason an organism is able to experience pain is because it possesses a state that is capable of carrying out the function of pain. It is a conceivable that the function of pain could be provided by a state of a material that is not physical at this point in time. This is a possibility that is entirely plausible. As a result of this, the statement that the functionalist substance dualism is accurate is one that can be considered plausible. They are of the opinion that it is highly conceivable that the actual physical states of the brain are responsible for carrying out the many causal functions that are associated with the various mental states. If functionalism is accurate, then there is a very good chance that a particular shape of the identity theory is authentic. Because of the close connection between the two ideas, this is the result.

Functionalism And Brain State

When attempting to characterise a state of the brain, it is possible to use the specific function of the brain. In the event that a person is experiencing pain, it is considered that they are going through an unpleasant brain state. Within a functionalist framework, it is possible to establish a wide variety of realisability, each of which can be considered distinct. It is possible, for instance, to characterise states of the brain such as hunger or pain by describing the functional duties that they have. Jaegwon Kim provides an instance of the notion of varied realisability in functional roles by using the mousetrap as an example. He is particularly well-known for his forceful criticism of non-reductive physicalism, which has made Jaegwon Kim one of the most influential contemporary philosophers of mind. Not only is the mouse trap analogy one of his numerous analogies, but it has also gained a lot of notoriety for its ability to convey the dilemma of multiple realisations and the pressure toward reduction. Kim is attempting to demonstrate, by means of this straightforward illustration, that higher-level characteristics might not possibly possess independent causal powers apart from their physical realisations.

It is a common argument among functionalists that mental states should be associated not with the components that make them up, but rather with the functions that they carry out. Pain, for example, is a mental state that may be defined by the manner in which it is caused, the manner in which it interacts with other states, and the manner in which it creates behaviour. Functionalists argue that mental states can be realised in a wide variety of materials, including human brains, animal brains, and even computers. This is due to the fact that different physical systems may execute the same function. This is the concept of having numerous aspects of reality.

In his statement, he does not refute the notion that several physical structures can carry out the same job. On the other hand, he raises the question of whether or not this demonstrates that higher-level qualities are irreducible or autonomous.

Additionally, Kim provides an explanation of the concept. In certain instances, a mousetrap will be equipped with a straightforward mechanism for activating the trap, but in other instances, it will be a wire cage that will close when sensors are activated. The incorporation of lasers or hydraulic power could be considered, depending on the specifics of the situation. Any device that is capable of capturing a mouse is regarded to be a mousetrap, regardless of its size, shape, or mechanism. This includes any and all devices that are capable of trapping mice. There is a connection between the states of the mind and the states of the brain, as the identity theory proposes as a possible explanation. There is a single instance of the category "C-fiber firing at the same time" that corresponds to each and every single case of the category "pain." The activation of the C-fibres and the experience of pain have a certain kind of identification with one another.

A concise explanation of the identity theory is that it claims that every type of mental state corresponds directly to a certain form of neurological condition. This is a statement that represents the theory. On the other hand, the identity theory does not require that every brain condition connects to a specific mental state in order to be valid. To be more specific, the states that are being taken into consideration are the mental states that the brain is currently experiencing. Furthermore, the identities should not be determined by only observing the individuals or by evaluating the meanings of the terminology that are pertinent to the situation. The brain state that is often created by the visual and aural cues of rain is precisely the same as the one that Thomas is convinced is occurring, which is that it is raining. A common occurrence is the coexistence of this brain state with other brain states that, among other things, result in the behaviour of retrieving an umbrella. Lewis posited that mental states are analogous to the

physical states that are present within us, and that these mental states serve particular functions that allow them to be considered causal. The upshot of this discovery is that the association between specific mental moods and specific physical conditions may vary from person to person within the same group. Also, this correlation may be different for different people. According to Lewis, the physical condition that corresponds to a specific mental state is dependent upon the causal roles that the state normally plays in the organism that possesses it. This is the point that Lewis is making. Pain is a unique form of brain state that is present in humans. All humans are capable of experiencing pain.

Functionalism And Six Features of Mental State

1There are particular states of the world that are the underlying cause of particular mental states. We have shown that the functional functions that are linked with various mental states are likely to be occupied by states of the brain. This is something that we have established. Therefore, the argument that certain mental states are caused by states of the world is only valid, according to the functionalist, if certain brain states are caused by states of the world

. 2. Various mental states are responsible for a variety of behaviours. Again, it is essential to bear in mind that it is quite possible that the states of the brain are the ones that occupy the functional functions that are specifically associated with mental states. This is something that should be kept in mind at all times. The argument that particular mental states are accountable for actions is only legitimate, according to the functionalist, if certain brain states are responsible for those actions. This is because the functionalist believes that action is a function of the brain.

3. It is possible to trace certain mental states back to the origin of other mental states. Given that mental states are states of the brain, the argument that certain mental states are responsible for other mental states is equal to asserting that certain brain states are responsible for other brain states. This is because mental states are states of the brain. Moreover, this is without a doubt the situation. Not only is it true that certain mental states are the cause of other mental states, but it is also true that the causal linkages between mental states can sometimes mirror the rational relations that exist between them. This is something that is true.

4. For certain mental states, it is feasible to be aware of them. Since the beginning of time, consciousness has always been a tremendous obstacle to overcome. It would appear that it is conceivable to think of a robot that possesses a central computer that is capable of occupying the functional role that is characteristic of pain, but does not experience it. This would be a remarkable achievement. It would appear that the functionalist approach faces a big obstacle in the form of consciousness if what you have stated is true.

5. There are things or events in the world that are related with certain mental states that people experience. From a theoretical standpoint, there are two distinct points of view regarding this topic. One of these, known as functional role semantics, is one of those that is quite at ease with the concept of functionalism. Alternatively, the other approach, which incorporates the causal theory of content, calls for additions to the underlying functionalist framework and, at the very least, some modifications. This is because the causal theory of content considers content to be a causal phenomenon.

6There is a continuous association between many different types of brain states and a wide variety of mental states, as researchers have discovered. It has been brought to our attention that it is not inconceivable that the functional functions that are distinctive of the various mental states be inhabited by brain states. Such a possibility has been brought to our attention. The functionalist viewpoint is compatible with the argument that there are systematic links between mental states and particular brain states. This is because the argument is based on the idea that there exist such ties.

Knock your Brain

1. Functionalism defines mental states mainly in terms of their:

A. Chemical composition

B. Physical location

C. Causal roles

D. Spiritual nature

2. Functionalism emerged largely as an alternative to:

A. Dualism only

B. Identity theory and behaviourism

C. Pragmatism

D. Existentialism

3. According to functionalism, a mental state is identified by:

A. Brain tissue

B. Input–output relations

C. Personal feelings only

D. Moral value

4. The idea that the same mental state can exist in different physical systems is called:

A. Supervenience

B. Eliminativism

C. Multiple realization

D. Parallelism

5. Which philosopher is strongly associated with functionalism?

A. B. F. Skinner

B. Hilary Putnam

C. René Descartes

D. David Hume

Unleash your potential

1.Elucidate the notion of functionalism within the philosophy of mind.

2.Examine the correlation between functionalism and artificial intelligence.

3.Elucidate the distinctions between functionalism, identity theory, and behaviourism.